Approximate minimax regret for conditionally similar tests

Andrés Aradillas Fernández[a]*

[a]Department of Economics, Massachusetts Institute of Technology
77 Massachusetts Avenue, Building E52-502
Cambridge, MA 02139, United States

November 2, 2026

Abstract

We construct approximately minimax-regret tests in a scalar Gaussian instrumental-variables experiment. This example shows how conditionally similar tests allow a novel approximation approach, allowing us to optimize weights over alternatives to minimize worst-case power regret. Stochastic mirror descent and averaging yield finite-iteration guarantees with exact conditional similarity. We establish bounds for exact and simulated powers, control discretization error in total variation, and illustrate the procedure numerically. This initial algorithm shows how existing nearly-optimal test methods can be extended to include composite alternate hypothesis sets, provided a conditioning statistic is obtainable.

Keywords: Conditional similarity; minimax regret; weak instruments; stochastic mirror descent.
JEL classification: C12, C19.

1 Introduction

Conditional similarity controls rejection probabilities in the presence of nuisance parameters but generally leaves many tests to choose from. We select among these tests by minimizing their largest power loss relative to the conditionally similar power envelope. This criterion is useful when instruments may be weak and no single test maximizes power at every alternative.

Our contribution is to use conditional size control to optimize power regret across alternatives. Elliott et al. (2015), henceforth EMW, fix a weighting distribution over alternatives; weighted average power then equals power against a single mixture distribution. Their numerical search concerns a least-favorable distribution under the null, and thus the algorithm seeks to find an optimal combination of tests that satisfy these size constraints simultaneously. Aradillas Fernández et al. (2025) develop stochastic mirror descent for the corresponding finite-null, single-alternative problem, thus attaining approximation guarantees not present in the algorithm of EMW. That formulation also accommodates a fixed alternative mixture representing a weighted-average-power (WAP) criterion (for a discussion, see Section 3.8 of Lehmann and Romano (2005)), restricting the problem to that of a composite null and a singleton alternative.

We instead impose conditional similarity within every best response. This removes the need to optimize null weights to enforce size and allows the iterative search to focus on alternatives at which power falls furthest below its conditionally similar envelope. How Aradillas Fernández et al. (2025) construct their algorithm is by utilizing the subgradient constructed from the Lagrangian, but by freeing up this size constraint, we are instead able to look at many combinations of alternate hypotheses. Weighted average power remains the

*Corresponding author. Email: aradilla@mit.edu.

best-response objective, but its weights are updated to solve a minimax-regret problem rather than specified in advance as the final power criterion. Our result concerns the conditionally similar class in the Gaussian experiment, instead of the unrestricted class of level-α tests, but this flexible method may lend itself to tractable results beyond Gaussian settings, which is accomplished in Aradillas Fernández et al. (2025) for the single-alternative version of the problem.

For a finite alternative grid, we combine conditional Neyman–Pearson best responses with exponential weight updates, the latter of which help preserve the geometry of the probability simplex. Averaging preserves exact conditional similarity and yields explicit regret bounds for exact and simulated power calculations.

Placing this in existing literature on conditionally similar tests, Andrews (2016) studies minimax regret within a class of conditional linear combination tests, while Montiel Olea (2020) characterizes admissible similar tests. We optimize over the full conditionally similar class here.

2 Setup and Algorithm

2.1 Testing problem

We observe a pair of reduced-form estimates satisfying

$$\begin{pmatrix} \hat{\delta} \\ \hat{\pi} \end{pmatrix} \sim N\left(\begin{pmatrix} \pi\beta \\ \pi \end{pmatrix}, \Sigma \right), \qquad \Sigma \text{ positive definite}, \tag{1}$$

where Σ is known and does not depend on (β, π). We test $H_0 : \beta = \beta_0$ at level $\alpha \in (0, 1)$, with unrestricted nuisance parameter $\pi \in \mathbb{R}$. Define

$$S = \hat{\delta} - \beta_0 \hat{\pi}, \qquad a = \mathrm{Cov}(\hat{\pi}, S)/\mathrm{Var}(S), \qquad U = \hat{\pi} - aS. \tag{2}$$

The transformation is invertible. Let $\sigma_S^2 = \mathrm{Var}(S)$ and $\sigma_U^2 = \mathrm{Var}(U)$. At $\theta = (\beta, \pi)$, the variables S and U are independent normals with means $m(\theta) = \pi(\beta - \beta_0)$ and $\tau(\theta) = \pi - am(\theta)$ and variances σ_S^2 and σ_U^2. Under the null, $S \mid U = u \sim N(0, \sigma_S^2)$, independently of π.

A randomized test is a measurable rejection probability $\varphi(s, u) \in [0, 1]$. Writing f_0 for the $N(0, \sigma_S^2)$ density, we consider

$$C_\alpha = \left\{ \varphi : \int \varphi(s, u) f_0(s)\, ds = \alpha \text{ for Lebesgue-a.e. } u \right\}. \tag{3}$$

Because the null density of U is everywhere positive, this restriction is equivalent to conditional similarity for every π. It implies unconditional rejection probability α.

For alternatives $\theta \in \Theta \subseteq \{(\beta, \pi) : \beta \neq \beta_0\}$, define the power envelope, regret, and minimax value by

$$b(\theta) = \sup_{\varphi \in C_\alpha} \mathbb{E}_\theta[\varphi], \qquad \mathcal{R}(\varphi, \theta) = b(\theta) - \mathbb{E}_\theta[\varphi], \qquad V_\Theta = \inf_{\varphi \in C_\alpha} \sup_{\theta \in \Theta} \mathcal{R}(\varphi, \theta). \tag{4}$$

We first take $\Theta = \Theta_I = \{\theta_1, \ldots, \theta_I\}$, with $I \geq 2$, and write $V_I = V_{\Theta_I}$ and $b_i = b(\theta_i)$.

2.2 Conditional best responses and weight updates

For weights w in the probability simplex Δ^{I-1}, let $g_w = \sum_i w_i g_i$, where g_i is the joint density of (S, U) at θ_i. The best response maximizes weighted average power:

$$\varphi_w \in \arg\max_{\varphi \in C_\alpha} \sum_{i=1}^{I} w_i \mathbb{E}_{\theta_i}[\varphi]. \tag{5}$$

Algorithm 1 Conditionally similar testing by stochastic mirror descent

Require: Grid Θ_I, size α, tolerance ϵ, failure probability δ, draws N.

Choose K and η according to (11).
Compute envelopes b_i using (7); set $w_{1,i} = 1/I$.
for $k = 1, \dots, K$ **do**
    Construct φ_k from (6), imposing conditional size α.
    Draw fresh $Y_{k,i,n} \sim g_i$ independently for all i, n.
    Set $\widehat{p}_{k,i} = N^{-1} \sum_n \varphi_k(Y_{k,i,n})$ for every i.
    Set $w_{k+1,i} \propto w_{k,i} \exp\{\eta(b_i - \widehat{p}_{k,i})\}$; normalize.
end for
return $\bar{\varphi}_K = K^{-1} \sum_k \varphi_k$.

The weights w are optimization variables for worst-case regret. Thus each best response uses a single alternative mixture, but the mixture changes across iterations as the algorithm reallocates weight toward high-regret alternatives. The constraint in (3) separates across u. Conditional Neyman–Pearson optimization therefore rejects for large values of

$$L_w(s,u) = \frac{g_w(s,u)}{f_0(s)} = \sum_{i=1}^{I} w_i f_U(u;\tau_i) \exp\left(\frac{m_i s}{\sigma_S^2} - \frac{m_i^2}{2\sigma_S^2}\right), \tag{6}$$

where $m_i = m(\theta_i)$, $\tau_i = \tau(\theta_i)$, and $f_U(\cdot;\tau_i)$ is the $N(\tau_i, \sigma_U^2)$ density. Specifically, $\varphi_w = \mathbf{1}\{L_w > c_w(u)\} + \gamma_w(u)\mathbf{1}\{L_w = c_w(u)\}$, with threshold and boundary randomization chosen to satisfy (3). For a single alternative, the optimal rule is one-sided in S, giving

$$b_i = \Phi\Big(\Phi^{-1}(\alpha) + |m_i|/\sigma_S\Big), \tag{7}$$

where Φ is the standard normal distribution function. When $m_i = 0$, the envelope equals α.

The function $F(w) = \sup_{\varphi \in C_\alpha} \sum_i w_i(\mathbb{E}_{\theta_i}[\varphi] - b_i)$ is convex, with subgradient $G_i(w) = \mathbb{E}_{\theta_i}[\varphi_w] - b_i$. Negative-entropy mirror descent thus increases weights in proportion to estimated regret (see Section 4.2 of Bubeck (2015), as well as Freund and Schapire (1999), after which this algorithm is modeled). Starting from $w_{1,i} = 1/I$, at iteration k we compute $\varphi_k = \varphi_{w_k}$ and set

$$\widehat{p}_{k,i} = \frac{1}{N}\sum_{n=1}^{N} \varphi_k(Y_{k,i,n}), \qquad \widehat{G}_{k,i} = \widehat{p}_{k,i} - b_i, \qquad w_{k+1,i} = \frac{w_{k,i}\exp(-\eta \widehat{G}_{k,i})}{\sum_j w_{k,j}\exp(-\eta \widehat{G}_{k,j})}. \tag{8}$$

Conditional on the preceding iterations, the draws $Y_{k,i,n} \sim g_i$ are independent across i and n and fresh at each iteration. Hence $\widehat{G}_k$ is conditionally unbiased. Exact-power updates replace $\widehat{p}_{k,i}$ by $\mathbb{E}_{\theta_i}\varphi_k$.

We return $\bar{\varphi}_K = K^{-1}\sum_{k=1}^{K} \varphi_k$, which could also be implemented by selecting an iteration uniformly and applying its test. Note that, conditional on simulation history, $\bar{\varphi}_K \in C_\alpha$ by construction. Thus simulation affects regret but does not affect conditional size when best responses are computed exactly. Algorithm 1 specifies the full implementation.

2.3 Algorithm guarantees

Proposition 1. *For the exact conditional best responses in* (5)*, the following statements hold for every $K \geq 1$ and $\eta > 0$.*

1. *With exact-power updates,*

$$\max_i \mathcal{R}(\bar{\varphi}_K, \theta_i) \leq V_I + \frac{\log I}{\eta K} + \frac{\eta}{8}. \tag{9}$$

2. *With the simulated updates in* (8)*, for any* $\delta \in (0,1)$*, with probability at least* $1-\delta$ *over the simulation draws,*

$$\max_i \mathcal{R}(\bar{\varphi}_K, \theta_i) \le V_I + \frac{\log I}{\eta K} + \frac{\eta}{2} + \sqrt{\frac{2\log((I+1)/\delta)}{NK}}. \tag{10}$$

With exact powers, $\eta = \sqrt{8\log I/K}$ gives error $\sqrt{\log I/(2K)}$. With simulated powers, an error tolerance $\epsilon > 0$ and failure probability δ are attained by

$$K = \left\lceil \epsilon^{-2}\left(\sqrt{2\log I} + \sqrt{\frac{2\log((I+1)/\delta)}{N}}\right)^2 \right\rceil, \qquad \eta = \sqrt{\frac{2\log I}{K}}. \tag{11}$$

The guarantees concern the average test, not the best response to average weights, which may underperform (e.g., see the discussion surrounding Algorith 1 of Aradillas Fernández et al. (2025)). Note that these results do not require a minimax interchange; weighted regret is bounded by V_I directly. Indeed, (13) also establishes $V_I = -\min_{w\in\Delta^{I-1}} F(w)$ by letting K increase, combining weak duality with the bound on average weighted regret. The continuity of F ensures attainment.

3 A numerical illustration

We set $\beta_0 = 0$, $\Sigma = I_2$, and $\alpha = 0.05$, so that $S = \hat{\delta}$ and $U = \hat{\pi}$. Our grid contains $\theta_1 = (-3, 0.5)$ and $\theta_2 = (1, 2)$, giving $(m_1, m_2) = (-1.5, 2)$ and $(\tau_1, \tau_2) = (0.5, 2)$. The corresponding power envelopes are 0.4424 and 0.6388. This deliberately small grid isolates the trade-off between power against negative and positive effects.

For these two alternatives, the likelihood ratio in (6) is strictly convex in s. The best response rejects in two tails. If $q_w(u)$ is its left-tail null probability, the critical values satisfy

$$\ell_w(u) = \Phi^{-1}(q_w(u)), \qquad h_w(u) = \Phi^{-1}(1-\alpha+q_w(u)), \qquad L_w(\ell_w(u), u) = L_w(h_w(u), u). \tag{12}$$

We solve the last equality by bisection over $q \in [10^{-14}, \alpha - 10^{-14}]$. The resulting rule has conditional size $q + (\alpha - q) = \alpha$ by construction. Larger values of u favor the second alternative and shift rejection probability toward the right tail.

We set the target excess regret to $\epsilon = 0.02$, the failure probability to $\delta = 0.05$, and the number of draws per alternative per iteration to $N = 512$. Equation (11), with $I = 2$, then selects $K = 4{,}251$ and $\eta = 0.01806$. Under exact conditional best responses, Proposition 1 gives probability at least 0.95 that the average test's maximum grid regret is at most $V_I + 0.02$. This guarantee concerns the stochastic mirror-descent updates; it does not cover numerical error in evaluating best responses. We evaluate power by integrating the conditional rejection probability with 320-node Gauss–Hermite quadrature. Doubling the quadrature order from 160 to 320 changes computed iteration-specific powers by less than 10^{-12}. The reported values come from one simulation run, rather than averages across repeated training runs, and use seed 20261102. The accompanying source records the implemented procedure and all simulation settings.

Table 1 shows that conditioning improves power at both alternatives relative to the equal-tail test. Figure 2 shows how the conditioning statistic shifts rejection probability between tails. Subsequent weight updates increase power at θ_1 while sacrificing some power at θ_2, reducing the largest regret. After 4,251 iterations, maximum regret is 0.0776, compared with 0.0819 initially and 0.1228 for the equal-tail test. Figure 1 reports a numerical dual lower bound of 0.0769, leaving a numerical gap of 0.00065. This diagnostic concerns the chosen grid and numerical best responses; it is not a uniform bound over unexamined alternatives. The theoretical tolerance is an excess over V_I, so it does not assert that the achieved maximum regret itself is below 0.02.

Table 1: Power and regret on the two-alternative grid

Test	Size	Power at θ_1	Power at θ_2	Max. regret
Equal-tail test	0.050	0.323	0.516	0.123
Initial best response	0.050	0.361	0.567	0.082
SMD average, 100 iterations	0.050	0.361	0.567	0.081
SMD average, 4251 iterations	0.050	0.365	0.563	0.078

Notes: The equal-tail test rejects when $|S| > \Phi^{-1}(0.975)$. The initial best response uses $w = (1/2, 1/2)$. SMD rows average all tests through the indicated iteration of the same run. Powers are evaluated by numerical integration; size follows from the tail allocation. Maximum regret is relative to the pointwise envelopes in (7).

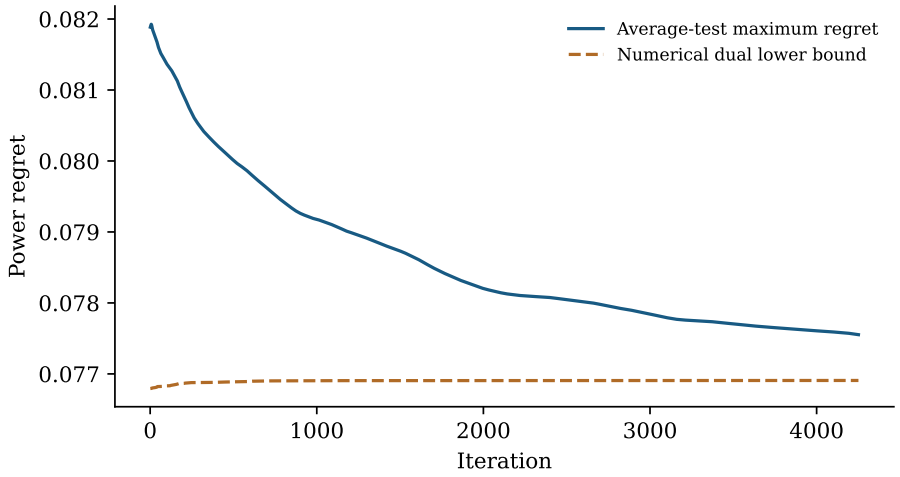


Figure 1: Regret along the SMD iterations. We report the maximum regret of the average test and the running maximum of $w_k^\top r_k$, evaluated by numerical integration. With exact best responses and integration, the latter bounds V_I from below.

4 Conclusion

Conditional similarity allows us to shift the numerical search from null distributions enforcing size to alternative distributions determining worst-case power regret. Relative to EMW's fixed WAP criterion, the alternative weights are chosen endogenously to balance regret within the conditionally similar class, allowing robustness to WAP misspecification. Conditional Neyman–Pearson best responses and stochastic mirror descent deliver finite-grid guarantees while preserving exact conditional size. Extensions beyond the grid require control of discretization error, and numerical implementations require accurate best responses.

Data and code availability

We use simulated data. The reproducible Python package, as well as instructions, are available in the public GitHub replication repository at version 1.1.0: `https://github.com/aaf101/SMD-for-Conditional-Similarity/tree/v1.1.0`.

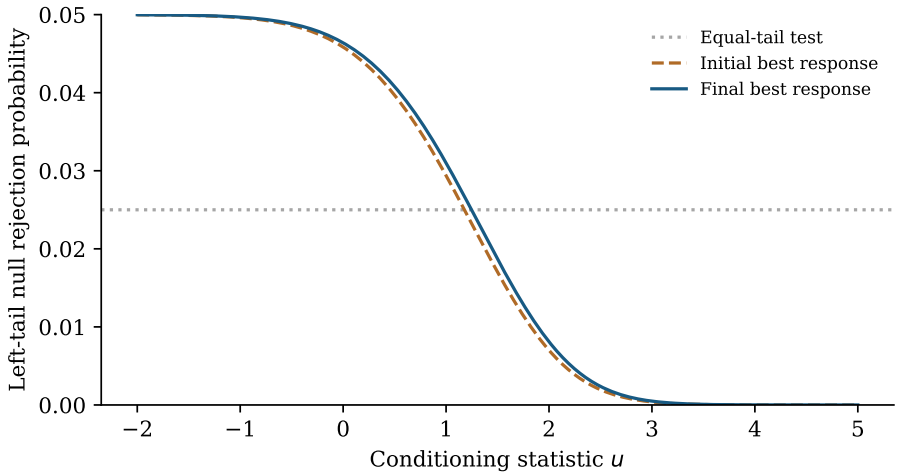


Figure 2: Conditional tail allocation. Left-tail null rejection probabilities for the initial and final individual best responses are compared with the equal-tail test. The returned test averages all iterates.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used GPT-6 to help with the writing, proof-checking, and code implementation in this paper. The author directed this work and subsequently reviewed and verified all output. The author takes full responsibility for the content of the article.

A Proofs of Main Results

A.1 Proof of Proposition 1

Write $r_{k,i} = \mathcal{R}(\varphi_k, \theta_i) \in [0,1]$. Best-response optimality gives $w_k^\top r_k = \inf_{\varphi \in C_\alpha} \sum_i w_{k,i} \mathcal{R}(\varphi, \theta_i) \le V_I$. The exponential-weights inequality for rewards in $[0,1]$ yields

$$\max_i \frac{1}{K} \sum_k r_{k,i} \le \frac{1}{K} \sum_k w_k^\top r_k + \frac{\log I}{\eta K} + \frac{\eta}{8}. \tag{13}$$

Linearity of power proves (9).

For simulated updates, $\widehat{G}_{k,i} \in [-1,1]$. The same inequality, now with range length two, gives, simultaneously for all i,

$$\frac{1}{K} \sum_k (r_{k,i} - w_k^\top r_k) \le \frac{\log I}{\eta K} + \frac{\eta}{2} + \frac{1}{K} \sum_k \Delta_{k,i} - \frac{1}{K} \sum_k w_k^\top \Delta_k, \tag{14}$$

where $\Delta_{k,i} = \widehat{G}_{k,i} - G_i(w_k)$. Conditional Hoeffding bounds and iteration over the simulation history imply $\Pr(K^{-1} \sum_k \Delta_{k,i} > x) \le \exp(-2NKx^2)$. The same bound holds for $-K^{-1} \sum_k w_k^\top \Delta_k$: conditional independence gives squared summand ranges totaling $\sum_i w_{k,i}^2 / N \le 1/N$ at each iteration. A union bound over these $I+1$ deviations, with $x = \sqrt{\log((I+1)/\delta)/(2NK)}$, bounds the two noise terms in (14) by $2x$. Using $w_k^\top r_k \le V_I$ and linearity proves (10).

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